Phishing Email Detection & Awareness Report

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Task 2- Phishing Email Detection & Awareness

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Introduction

Phishing is a type of cyber-attack in which attackers impersonate legitimate organizations or trusted individuals to trick users into revealing sensitive information such as passwords, banking details, or one-time PINs (OTPs). These attacks are typically delivered through email and are designed to appear urgent or important to manipulate victims into taking immediate action.

Phishing is dangerous because it targets human behaviour rather than technical system vulnerabilities. Even well-secured systems can be compromised if a user unknowingly clicks a malicious link, downloads an infected attachment, or submits login credentials on a fake website. Successful phishing attacks can lead to financial loss, identity theft, data breaches, and reputational damage for organizations.

The purpose of this report is to analyse phishing email samples, identify common phishing indicators, classify the level of risk, and provide practical awareness guidelines to help employees recognize and prevent phishing attacks. This report is designed to support cybersecurity awareness and reduce the likelihood of successful social engineering attacks within an organization.

1. Email sample 1

Email summary

This email claims that they have identified suspicious activities in my account and that I should verify my details to avoid suspension of my account.

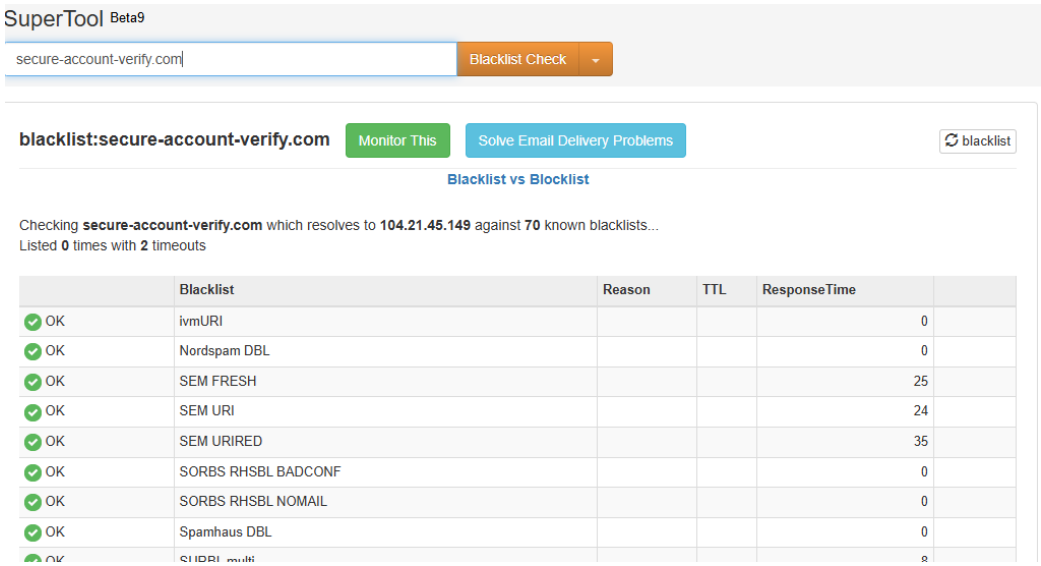
Identified phishing indicators:

- Urgency language
- Generic greeting
- Suspicious URL
- Threat-based messaging
- No official branding

Risk classification

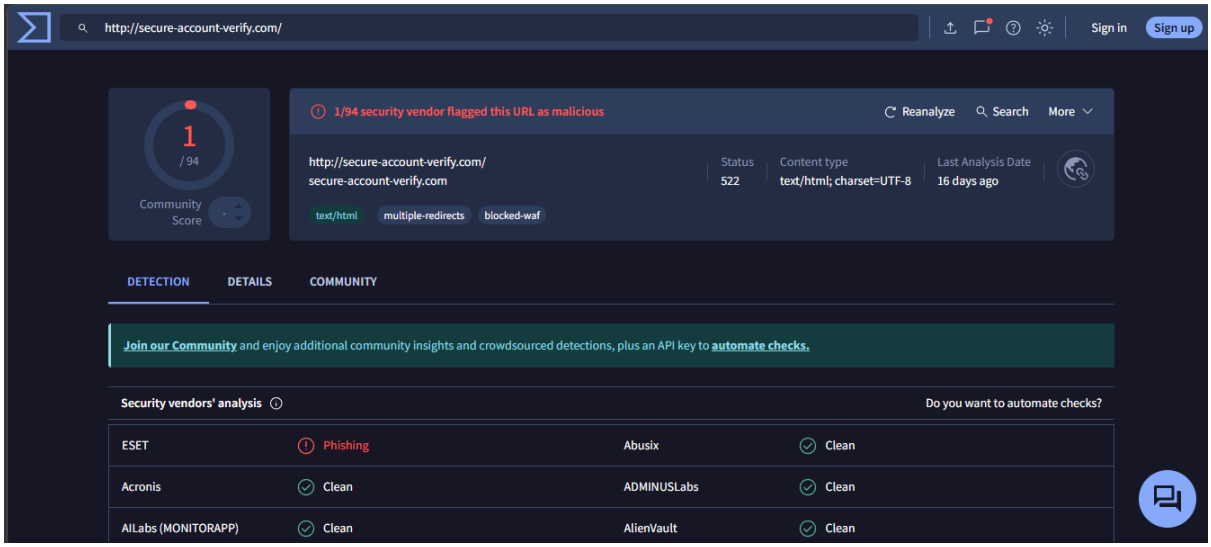
Risk level: High- Confirmed phishing email

Domain and infrastructure analysis



	Blacklist	Reason	TTL	ResponseTime
OK	ivmURI			0
OK	Nordspam DBL			0
OK	SEM FRESH			25
OK	SEM URI			24
OK	SEM URIRED			35
OK	SORBS RHSBL BADCONF			0
OK	SORBS RHSBL NOMAIL			0
OK	Spamhaus DBL			0

Figure 1 Blacklist check on MXToolbox



The domain *secure-account-verify.com* was analysed using MXToolbox Blacklist Check.

Findings:

- Resolved IP Address: *104.21.45.149*
- Checked Against: 70 blacklist databases
- Blacklist Status: Not listed
- Timeouts: 2 blacklist services did not respond

Interpretation of Results

Although the domain is not currently listed on any blacklist databases, this does not confirm that the domain is legitimate or safe.

Phishing domains are often:

- Newly registered
- Active for a short period
- Removed before detection systems update
- Frequently rotated by attackers

The structure of the domain name ("secure-account-verify") strongly resembles impersonation tactics commonly used in phishing campaigns. Legitimate companies typically use official branded domains rather than generic security-related phrases.

Risk Assessment Based on Domain Analysis

Even though the domain is not blacklisted, when combined with:

- Urgent fear-based messaging
- Generic greeting
- Suspicious URL structure
- Lack of official branding

The overall risk remains HIGH

2. How this phishing attack works

- Attacker sends fake alert email.
- Victim clicks the link.
- Fake website captures victims' credentials.
- Attackers' login into the real account.
- Financial or data loss occurs.

3. Phishing prevention guidelines

Employees should not:

- Click urgent email links.
- Share OTP codes
- Download unexpected attachments
- Trust unknown senders

Employees should:

- Hover over links
- Verify sender very carefully
- Report suspicious emails
- Contact company directly from their company website

4. Risk classification table

Indicator	Present?
Urgency tactic	Yes
Suspicious domain	Yes
SPF Failure	Yes
Generic greeting	Yes
Overall risk	HIGH

Conclusion

Phishing attacks remain one of the most common and effective forms of cybercrime because they exploit human behaviour rather than technical vulnerabilities. Attackers use

psychological tactics such as urgency, fear, and impersonation to trick users into clicking malicious links or revealing sensitive information.

In this report, a phishing email sample was analysed to identify common indicators used in phishing attacks. Several warning signs were detected, including a suspicious domain name, urgency-based messaging, and the use of a generic greeting. The domain reputation analysis showed that the domain was not yet blacklisted, demonstrating that newly created phishing domains may not always be immediately detected by security tools.

The overall risk classification of the analysed email was determined to be High Risk – Phishing Attempt. This highlights the importance of combining technical analysis with user awareness when detecting phishing threats.

Organizations can significantly reduce the success rate of phishing attacks by implementing regular cybersecurity awareness training, encouraging employees to verify suspicious emails, and promoting safe email practices. Educating users to recognize phishing indicators is one of the most effective defences against social engineering attacks.